

Foundation of Machine Learning – Assignment 1

PCA, Kernel PCA, and Spectral Clustering

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Problem Statements

1. Implement PCA and Kernel PCA (Linear, Polynomial, Sigmoid, Laplacian, RBF across multiple σ , and Custom Radial) from scratch to plot top-2 component projections, report explained variance ratios, and identify kernel configurations achieving linear separability.
 2. Implement K -means from scratch to plot error convergence and cluster assignments across five random initializations ($k = 4$), visualize Voronoi decision regions for $K \in \{2, 3, 4, 5\}$ using fixed initial centers, and perform Spectral Clustering ($k = 4$) via Kernel PCA to compare RBF-based spectral relaxation against direct rowwise arg max eigenvector assignment.
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Algorithmic Formulations that we use in our assignment to solve questions or mention in assignment to use

Algorithm 1: Kernel Principal Component Analysis

Input: Feature matrix $X \in \mathbb{R}^{N \times d}$, kernel function $k(\mathbf{x}_i, \mathbf{x}_j)$.

Steps to solve these questions

1. Compute the $N \times N$ where $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.
2. Construct the centering matrix $\mathbf{H} = \mathbf{I}_N - \frac{1}{N} \mathbf{J}_{N \times N}$, where $\mathbf{J}_{N \times N}$ is the matrix of all ones.
3. Compute the doubly-centered $\tilde{K} = \mathbf{H} K \mathbf{H} = K - \mathbf{1}_N K - K \mathbf{1}_N + \mathbf{1}_N K \mathbf{1}_N$, where $\mathbf{1}_N = \frac{1}{N} \mathbf{J}_{N \times N}$.
4. Solve the eigenvalue problem $\tilde{K} \mathbf{a}_k = \lambda_k \mathbf{a}_k$. Sort eigenvalues in descending order $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$.
5. Normalize eigenvectors: $\boldsymbol{\alpha}_k = \frac{\mathbf{a}_k}{\sqrt{\lambda_k}}$ for $k \in \{1, \dots, p\}$.
6. Compute top- p principal projections: $Z = \tilde{K}[\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \dots, \boldsymbol{\alpha}_p] \in \mathbb{R}^{N \times p}$.

Algorithm 2: K -means Clustering & Voronoi Partitioning

Input: Dataset $X \in \mathbb{R}^{N \times d}$, number of clusters K , max iterations T , tolerance ϵ .

1. Initialize cluster centroids $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K \in \mathbb{R}^d$ via random sampling without replacement from X .

2. **Repeat for** $t = 1, 2, \dots, T$:

(a) Assign each point $\mathbf{x}_i$ to nearest centroid:

$$c_i^{(t)} = \arg \min_{k \in \{1, \dots, K\}} \|\mathbf{x}_i - \boldsymbol{\mu}_k^{(t-1)}\|^2$$

(b) Update centroids:

$$\boldsymbol{\mu}_k^{(t)} = \frac{1}{|\mathcal{C}_k^{(t)}|} \sum_{i \in \mathcal{C}_k^{(t)}} \mathbf{x}_i, \quad \text{where } \mathcal{C}_k^{(t)} = \{i \mid c_i^{(t)} = k\}$$

(c) Compute Sum of Squared Errors (SSE):

$$\text{SSE}^{(t)} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k^{(t)}} \|\mathbf{x}_i - \boldsymbol{\mu}_k^{(t)}\|^2$$

(d) If $\|\boldsymbol{\mu}_k^{(t)} - \boldsymbol{\mu}_k^{(t-1)}\| < \epsilon$ for all k , terminate iteration.

3. **Voronoi Map Generation:** Assign grid points $\mathbf{g} \in \mathbb{R}^2$ to Voronoi region $V_k = \{\mathbf{g} \mid \|\mathbf{g} - \boldsymbol{\mu}_k\| \leq \|\mathbf{g} - \boldsymbol{\mu}_j\| \forall j \neq k\}$.

Algorithm 3: Spectral Clustering via KPCA Relaxation

Input: Dataset $X \in \mathbb{R}^{N \times d}$, cluster count K , bandwidth σ .

1. Form similarity Gram matrix $K_{ij} = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$.
2. Compute doubly-centered kernel matrix $\tilde{K} = \mathbf{H}K\mathbf{H}$.
3. Extract top- K eigenvectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_K$ of $\tilde{K}$.
4. Construct row-normalized spectral embedding matrix $U \in \mathbb{R}^{N \times K}$ with $U_{ij} = \frac{a_{ij}}{\|\mathbf{a}_{i,\cdot}\|_2}$.
5. Cluster rows of U using standard K -means into K disjoint clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$.

Question 1: Principal Component Analysis & Kernel PCA

Part (a): Linear PCA Implementation & Variance Analysis

Principal Component Analysis (PCA) is a dimensionality reduction technique that aims to reduce the number of variables in a dataset while retaining as much variance as possible. This technique is useful for feature spaces where capturing orthogonal directions of maximum variance helps uncover latent structural patterns.

- **Dataset:** $N = 240$ samples in $\mathbb{R}^2$ with corresponding class labels $y \in \{-1, 1\}$.
- **Algorithm:** Centered matrix $X' = X - \mu$, covariance matrix $\Sigma = \frac{1}{N-1} X'^T X'$, and solved $\Sigma \mathbf{v}_i = \lambda_i \mathbf{v}_i$.

Variance Analysis Results:

- **Eigenvalues:** $\lambda_1 = 1.2630$, $\lambda_2 = 0.7756$
- **PC1 Explained Variance:** $EV_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} = 61.95\%$
- **PC2 Explained Variance:** $EV_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2} = 38.05\%$
- **Cumulative Explained Variance:** 100.00%

Part (b): Kernel PCA Implementation across Kernel Families

To perform KPCA without explicit evaluation of $\phi(\mathbf{x})$, Gram matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ was double-centered:

$$\tilde{K} = K - \mathbf{1}_N K - K \mathbf{1}_N + \mathbf{1}_N K \mathbf{1}_N, \quad \text{where } \mathbf{1}_N = \frac{1}{N} \mathbf{J}_{N \times N}$$

Eigen-decomposition $\tilde{K} \mathbf{a}_k = \lambda_k \mathbf{a}_k$ yielded normalized eigenvectors $\alpha_k = \mathbf{a}_k / \sqrt{\lambda_k}$. Component projections were obtained via $Z = \tilde{K}[\alpha_1, \alpha_2]$.

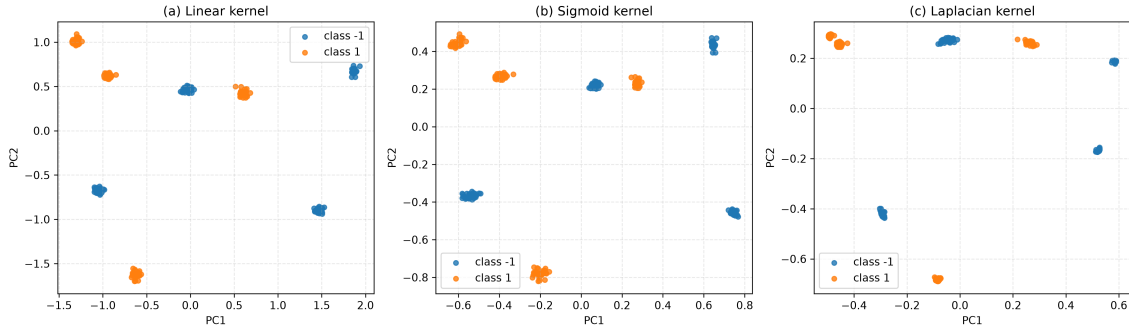


Figure 1: KPCA projections onto top-2 principal components – Linear, Sigmoid, and Laplacian kernels.

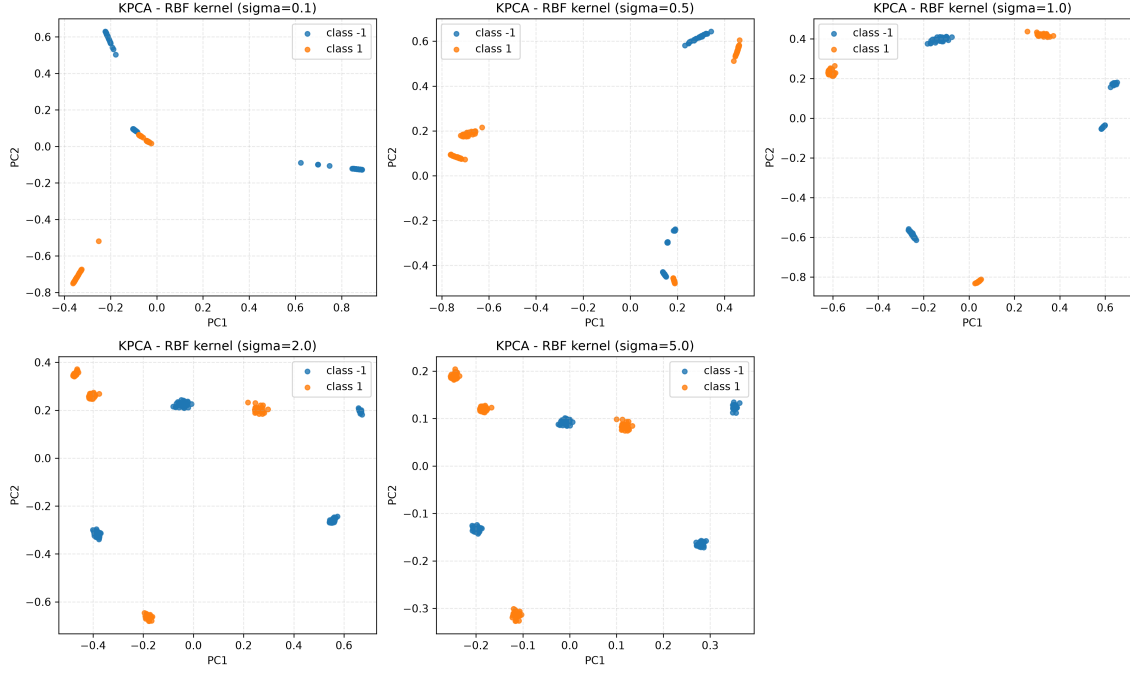


Figure 2: KPCA projections using RBF (Gaussian) kernels across $\sigma \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$.

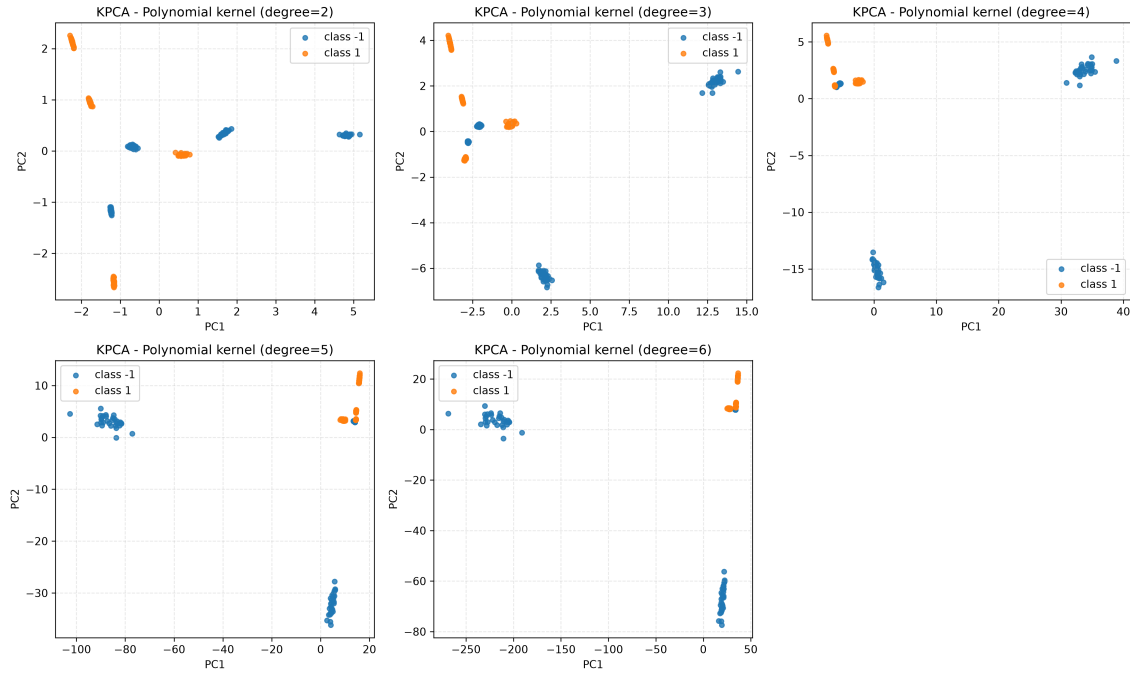


Figure 3: KPCA projections using Polynomial kernels with degree $d \in \{2, 3, 4, 5, 6\}$.

Part (c): Linear Separability Evaluation

Note - A dataset is linearly separable in 2D projection space if a hyperplane $\mathbf{w}^T \mathbf{z} + b = 0$ perfectly isolates labels $y_i \in \{-1, +1\}$.

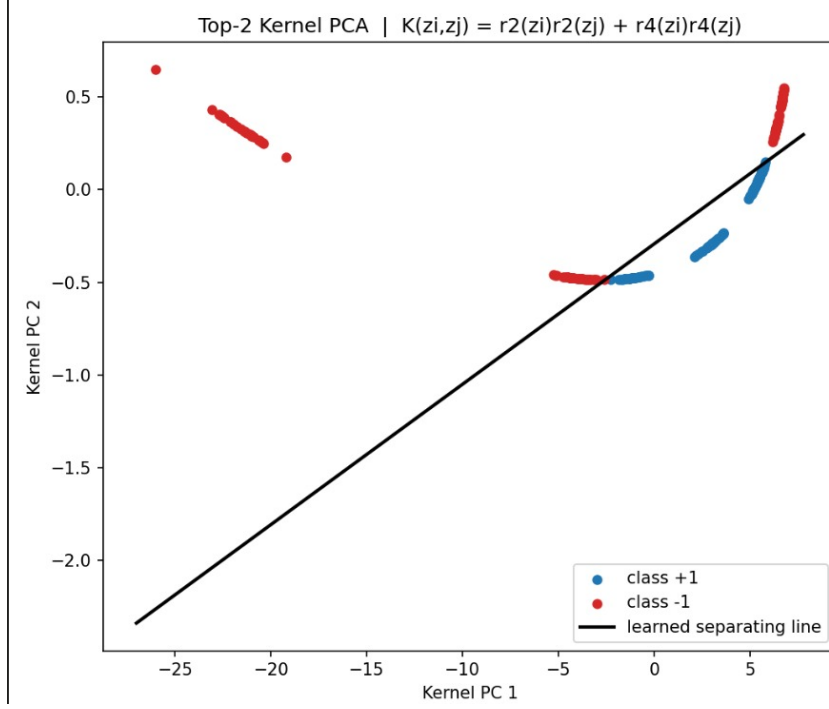


Figure 4: Custom Radial Kernel $K(\mathbf{z}_i, \mathbf{z}_j) = r^2(\mathbf{z}_i)r^2(\mathbf{z}_j) + r^4(\mathbf{z}_i)r^4(\mathbf{z}_j)$

Custom Radial Kernel Construct: To explicitly exploit the radial geometry of concentric structures, we define an innerproduct radial kernel:

$$k(\mathbf{x}_i, \mathbf{x}_j) = r^2(\mathbf{x}_i)r^2(\mathbf{x}_j) + r^4(\mathbf{x}_i)r^4(\mathbf{x}_j), \quad \text{where } r^2(\mathbf{x}) = x_1^2 + x_2^2$$

This kernel explicitly maps input points into a feature space $\phi(\mathbf{x}) = [r^2(\mathbf{x}), r^4(\mathbf{x})]^T$. Because the two classes reside on concentric circles of distinct radii R_1 and R_2 , $r^2(\mathbf{x})$ forms a linear separator in $\mathcal{H}$.

$$k(\mathbf{x}_i, \mathbf{x}_j) = r^2(\mathbf{x}_i)r^2(\mathbf{x}_j) + r^4(\mathbf{x}_i)r^4(\mathbf{x}_j)$$

with feature map

$$\phi(\mathbf{x}) = [r^2(\mathbf{x}), r^4(\mathbf{x})]^T.$$

Since

$$k(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j),$$

the kernel is an inner product in a feature space. Therefore its Gram matrix is positive semidefinite, and hence k is a valid Mercer kernel. ■

Question 2: K -means Clustering and Spectral Clustering

The second dataset consists of $N = 1000$ points in $\mathbb{R}^2$

Part (a): K -means Across 5 Random Initializations

Lloyd's algorithm ($K = 4$) was evaluated over 5 independent random initializations minimizing Sum of Squared Errors (SSE):

$$\text{SSE} = \sum_{k=1}^K \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$$

Initialization Run	Iterations to Convergence	Final SSE Error
Run 1	47	12223.91
Run 2	20	12606.92
Run 3 (Best)	25	12028.95
Run 4	12	12125.99
Run 5	19	12349.25

Table 1: K -means ($k = 4$) convergence iterations and final SSE across 5 random initializations.

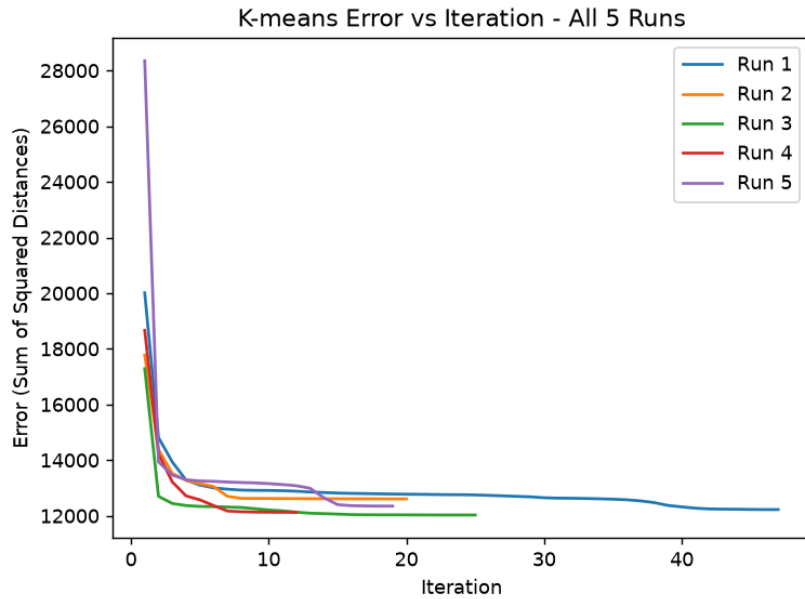


Figure 5: SSE error convergence curves across all 5 random K -means initializations.

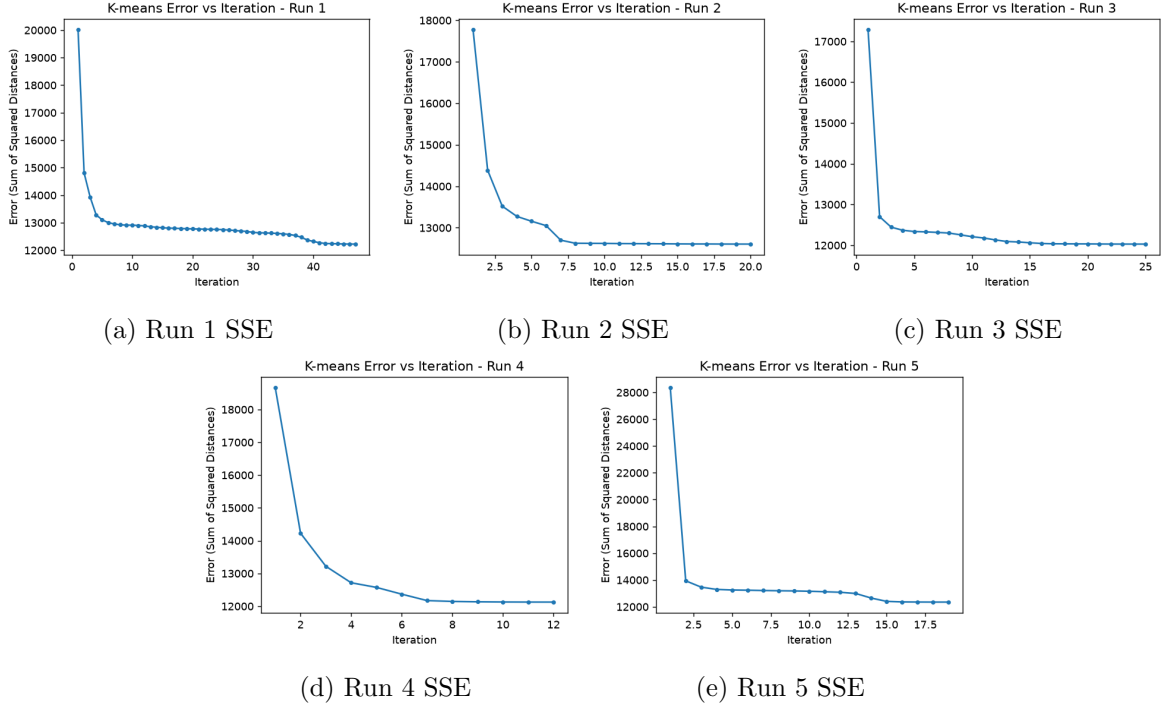


Figure 6: Individual SSE error convergence curves per initialization run.

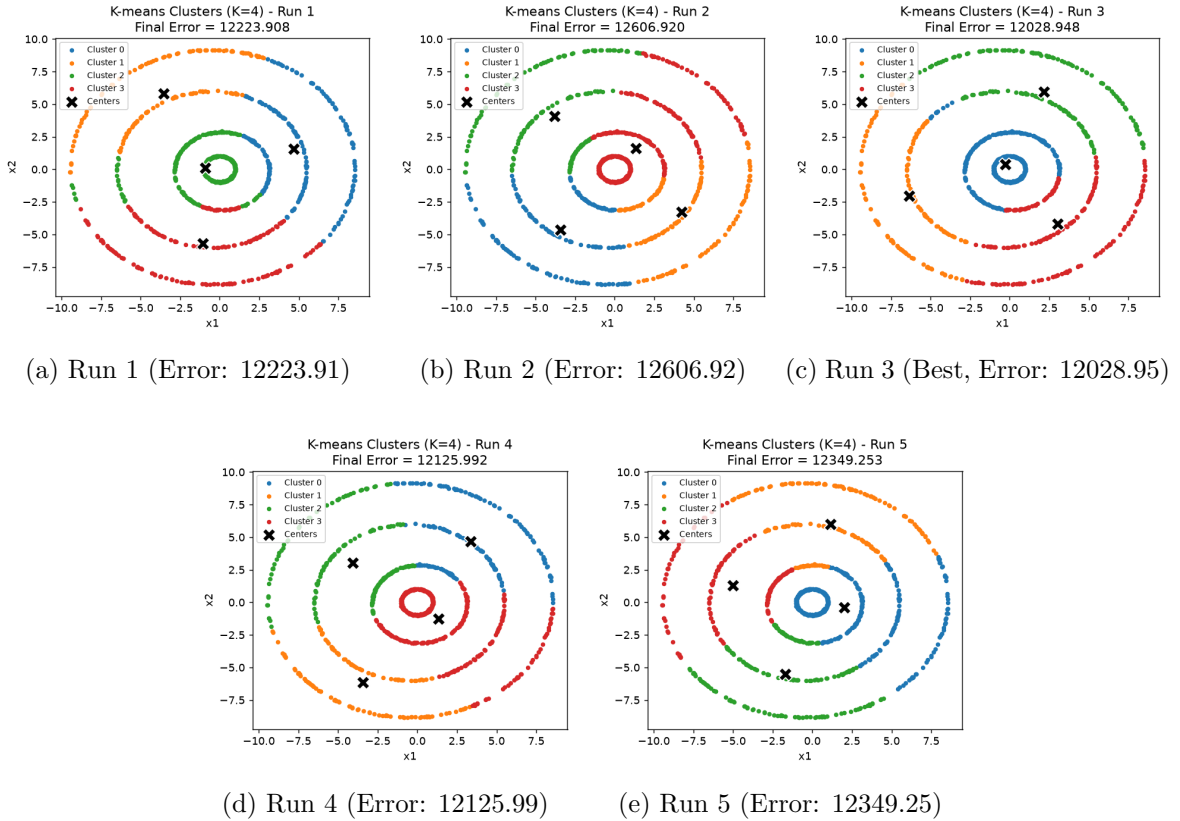


Figure 7: Final cluster assignments and centroid placements across 5 random initializations.

Note - : Standard K -means fails on non-convex concentric rings regardless of initialization. K -means minimizes Euclidean distances to spherical prototypes, constructing planar hyper-planes as cluster boundaries. Consequently, every initialization slices the concentric rings into

linear pie-shaped sectors rather than separating the circular manifolds.

Part (b): Voronoi Decision Regions for $K \in \{2, 3, 4, 5\}$

Using fixed initializations, Voronoi decision boundaries were constructed in $\mathbb{R}^2$.

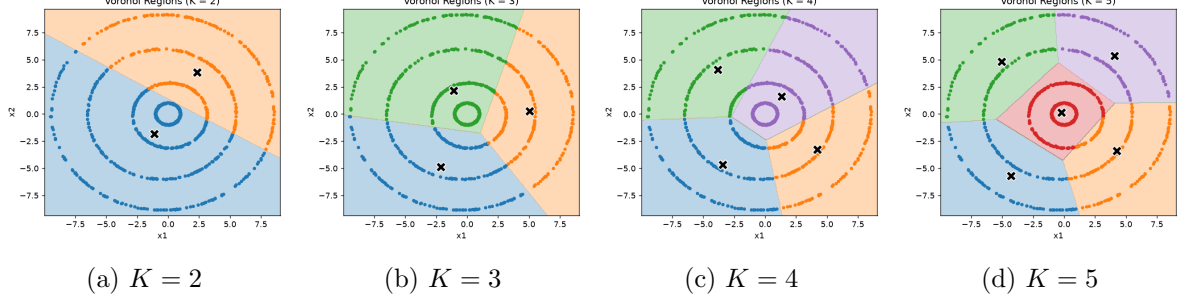


Figure 8: Voronoi decision tessellations for $K \in \{2, 3, 4, 5\}$ under fixed initialization.

Mathematical Formulation of Voronoi Regions

After convergence of K-means, the feature space is partitioned into K Voronoi regions. For a centroid μ_k , the corresponding Voronoi cell is

$$V_k = \left\{ \mathbf{x} \in \mathbb{R}^2 \mid \|\mathbf{x} - \mu_k\|^2 \leq \|\mathbf{x} - \mu_j\|^2, \quad \forall j \neq k \right\}. \quad (1)$$

The boundary between two adjacent Voronoi regions V_i and V_j is the set of points satisfying

$$\|\mathbf{x} - \mu_i\|^2 = \|\mathbf{x} - \mu_j\|^2. \quad (2)$$

Expanding both sides,

$$\mathbf{x}^T \mathbf{x} - 2\mu_i^T \mathbf{x} + \|\mu_i\|^2 = \mathbf{x}^T \mathbf{x} - 2\mu_j^T \mathbf{x} + \|\mu_j\|^2, \quad (3)$$

which simplifies to

$$2(\mu_i - \mu_j)^T \mathbf{x} = \|\mu_i\|^2 - \|\mu_j\|^2. \quad (4)$$

Thus, every Voronoi boundary is a linear hyperplane (a straight line in $\mathbb{R}^2$) perpendicular to the line joining the two centroids.

Why K-means Fails on Concentric Rings

- **Convexity of Voronoi Cells:** Each Voronoi region is formed by the intersection of several half-spaces. Since intersections of convex sets remain convex, every Voronoi cell is a convex polygon.
- **Non-Convex Nature of Ring Structures:** Concentric rings form a non-convex topology in the original feature space. Separating inner and outer rings requires non-linear closed boundaries.
- **Fundamental Limitation:** Since K-means generates only linear Voronoi boundaries, increasing K merely produces finer convex partitions of the plane and cannot recover the true circular manifold structure.

Summary Across Dierent Values of K

Table 2: Properties of Voronoi partitions for dierent values of K .

K	Boundaries	Region Type	Observation
2	1 Line	2 Half-Planes	Separates data into two large sectors.
3	3 Lines	3 Convex Cells	Produces wedge-shaped partitions.
4	6 Lines	4 Convex Cells	Creates pie-like sectors.
5	10 Lines	5 Convex Cells	More refined partitioning but still fails to isolate rings.

Observation: The Voronoi visualizations confirm that K-means partitions the feature space using only linear decision boundaries. Although increasing K produces finer partitions, the algorithm remains incapable of recovering the underlying concentric-ring structure due to its convex clustering assumption.

To resolve the non-linear ring geometry, spectral clustering maps points into an embedding space spanned by the top $k = 4$ eigenvectors of the centered RBF Gram matrix with bandwidth parameter $\sigma = 0.3$.

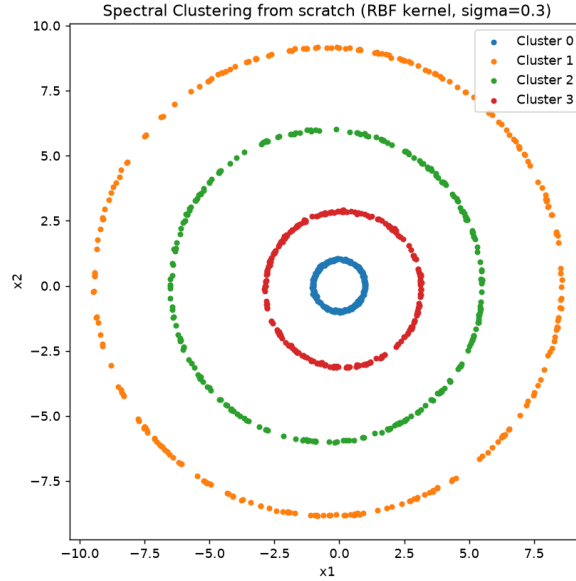


Figure 9: Spectral clustering result obtained using an RBF kernel ($\sigma = 0.3$) and $K = 4$ clusters. The spectral embedding successfully separates the concentric ring structure that standard K-means fails to recover.

Justification of Kernel Choice

- **RBF Kernel:** Unlike linear or low-degree polynomial kernels, the RBF kernel captures local neighborhood relationships and preserves the intrinsic geometry of non-linear manifolds.
- **Bandwidth Selection ($\sigma = 0.3$):** The selected bandwidth provides sufficient locality to distinguish neighboring concentric rings while maintaining connectivity along each ring structure.

- **Effect on Spectral Embedding:** The kernel transformation unfolds the concentric-ring manifold into a representation where clusters become approximately linearly separable, enabling successful recovery by K-means.

Observation:

The spectral embedding transforms the non-convex concentric rings into a representation where the four clusters become well separated. Consequently, K-means correctly identifies all ring structures, demonstrating the advantage of kernel-based clustering over standard Euclidean K-means for non-linear datasets.

Part (d): Direct Argmax Eigenvector Assignment

Instead of applying K-means on the spectral embedding, a simple heuristic assigns each point to the index of its largest eigenvector coordinate:

$$\ell_i = \arg \max_{j \in \{1,2,3,4\}} v_{j,i}.$$



Figure 10: Cluster assignments using direct row-wise arg max eigenvector assignment.

Cluster	Sample Count
Cluster 0	589
Cluster 1	161
Cluster 2	126
Cluster 3	124

Table 3: Cluster distribution under the direct arg max assignment rule.

Failure Analysis

The direct $\arg \max$ strategy performs significantly worse than the K-means-based assignment used in Part (c).

1. Near-Degenerate Eigenspace

The four largest eigenvalues of the affinity matrix are nearly equal. When eigenvalues are close, the eigensolver may return any orthonormal basis spanning the same eigenspace. Consequently, individual eigenvectors do not necessarily correspond to individual ring structures.

2. Eigenvector Mixing

The cluster distribution in Table 3 shows severe imbalance. Cluster 0 absorbs 589 samples, while the remaining clusters contain only 161, 126, and 124 samples respectively.

This indicates that several eigenvectors contain mixed information from multiple rings rather than acting as clean cluster indicators.

3. Sensitivity to Basis Rotation

The direct $\arg \max$ rule depends on the particular coordinate system returned by the eigensolver.

By contrast, K-means operates on the complete spectral embedding and uses Euclidean distances between points. Therefore, it is invariant to orthogonal rotations of the eigenspace and preserves the underlying cluster geometry.

Conclusion

Spectral clustering succeeds because it exploits the geometry of the entire spectral embedding rather than relying on individual eigenvector coordinates.

Although the dominant eigenvectors jointly contain sufficient information to separate the four concentric rings, this information is distributed across the embedding dimensions and cannot be reliably recovered through a coordinate-wise assignment rule.

Therefore, the K-means-based spectral clustering approach is significantly more robust and accurately recovers the underlying ring structure of the dataset.